



Cambridge IGCSE™

BIOLOGY

0610/62

Paper 6 Alternative to Practical

October/November 2020

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2020 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **9** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

mark scheme abbreviations

- ; separates marking points
- / alternative responses for the same marking point
- **not** do not allow
- **allow** accept the response
- **ecf** error carried forward
- **AW** alternative wording (where responses vary more than usual)
- **AVP** alternative valid point
- underline actual word given must be used by candidate (grammatical variants excepted)
- max indicates the maximum number of marks that can be given.
- **ora** or reverse argument
- () the word / phrase in brackets is not required but sets the context
- **Any [number] of:** accept the [number] of valid responses
- **R** reject
- **A** accept (for answers correctly cued by the question, or guidance for examiners)
- **I** ignore as irrelevant

Question	Answer	Marks	Guidance									
1(a)(i)	table with a line between headings and data <u>and</u> a minimum of two columns or rows ; correct headings for column / rows given ; record of four correct heights ± 1 mm ; a horizontal or vertical line draw on at least two cups on Fig. 1.1 indicating where measurements were taken ;	4	<table><tr><td>cup</td><td>start height / mm</td><td>end height / mm</td></tr><tr><td>cool</td><td>36 (35 – 37)</td><td>45 (44 – 46)</td></tr><tr><td>warm</td><td>39 (38 – 40)</td><td>67 (66 – 68)</td></tr></table>	cup	start height / mm	end height / mm	cool	36 (35 – 37)	45 (44 – 46)	warm	39 (38 – 40)	67 (66 – 68)
cup	start height / mm	end height / mm										
cool	36 (35 – 37)	45 (44 – 46)										
warm	39 (38 – 40)	67 (66 – 68)										
1(a)(ii)	C = 9 mm / 0.9 cm W = 28 mm / 2.8 cm	1	ecf for incorrect initial measurements									
1(a)(iii)	at high(er) temperatures, the dough rises more / more carbon dioxide is produced / AW / ora	1	ecf from candidate's data									
1(a)(iv)	temperature ;	1										
1(a)(v)	(maximum) height (of dough) / distance from bottom of cup to top of dough ;	1										

PUBLISHED

Question	Answer	Marks	Guidance
1(b)(i)	<i>error</i> the dough was not accurately divided into two / AW ; <i>improvement</i> measure the mass / use the same mass / use a balance / AW (to ensure both dough sections have equal mass) ;	2	
1(b)(ii)	idea of the height of the dough at the beginning was not identical in both cups ;	1	
1(c)	add iodine (solution / reagent) ; blue-black (colour) ;	2	
1(d)(i)	hydrogencarbonate (indicator) ; yellow / orange (colour) ; or limewater ; cloudy / white / milky / chalky ;	2	

PUBLISHED

Question	Answer	Marks	Guidance
1(d)(ii)	<i>independent variable:</i> 1 at least two different masses or amounts of sugar used / masses stated ; <i>given method:</i> 2 putting dough, into same sized cups / containers ; 3 measuring height / volume of dough ; <i>new method:</i> 4 described method of maintaining temperature ; 5 a detail of using the yeast suspension ; 6 measurement method ; e.g. gas syringe / delivery tube and water displacement / delivery tube and counting bubbles / measuring height of foam formed / bubbling gas through (named) indicator <i>dependent variable:</i> 7 measure volume of gas (carbon dioxide) produced in set time / measure time to produce set volume of gas / measure time for positive indicator result ; <i>variables kept constant:</i> 8 and 9 any two from: ;; temperature for all samples species / age / source, of yeast type of sugar / sugar from same sample mass of yeast / concentration of yeast (suspension) / volume of yeast (suspension) pH / use of buffer <i>other:</i> 10 two or more repeats / replicates ; 11 wear goggles / eye protection ; 12 AVP ; e.g. control experiments described	6	

Question	Answer	Marks	Guidance
2(a)(i)	single clear outline and no shading ; cell at least 10 cm from left to right ; <i>details:</i> two clusters of chloroplasts (minimum of three in left cluster and four in right cluster) ; two 'protrusions' in cell wall at cell junctions top and bottom ;	4	
2(a)(ii)	(length of PQ) = 94 ± 1 (mm) ; 0.078 / 0.08 mm ;;	3	MP1 correct measurements MP2 correct calculation MP3 correct unit
2(b)(i)	<u>103</u> (bubbles per minute) ;	1	
2(b)(ii)	axes labelled with units ; even scale and with plots using at least half the grid in one direction ; six points plotted accurately $\pm$ half a small square ; suitable line drawn ;	4	
2(b)(iii)	43 ± 1 (bubbles per minute) ;	1	A candidate's line at 170 cm $\pm$ $\frac{1}{2}$ small square
2(b)(iv)	<i>as plant moves further away from the light: ora</i> number of bubbles remains constant and then decreases ; change in rate, after 100–139 cm / after the first two readings (either distance or bubbles) / after the first three readings / AW ;	2	
2(b)(v)	idea of plant equilibrating to the new conditions / AW ;	1	

PUBLISHED

Question	Answer	Marks	Guidance
2(b)(vi)	39 (%) ;;;	3	MP1 selection of correct data MP2 correct calculation MP3 correct rounding to a whole number